EQUIVALENCE OF THREE LOCAL FORMS OF LOCAL EQUIDIMENSIONALITY FOR THE RELATIVE DIMENSION FUNCTION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as three spans, lines 8-12 (the verbatim package together with the project's own reference, slogan and history display-environment definitions, including the author's commented-out older form of the reference environment), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). The lines 8-12 span is required because a counted statement or its surrounding text may use those display environments, which the project typesets as comments. Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Editorial note. The author's footnote attached to the Cohen-Macaulay clause of the statement (source lines 8294-8296) is retained verbatim inside the counted statement.

Allowed standard prerequisites (editorial).

- A normal locally Noetherian scheme is a disjoint union of integral schemes. Properties, label properties-lemma-normal-locally-Noetherian, invoked at proof line 8307. Gives the normal case, since on each integral piece the dimension function is constant at generic points.
- Uniqueness up to a locally constant function for dimension functions, and the Cohen-Macaulay example. Topology, label topology-lemma-dimension-function-unique (proof line 8313); Example example-CM-irreducible (proof line 8311). Shows that the negative of the dimension of the local ring is a dimension function which vanishes at generic points, so that the relative dimension function differs from it by a locally constant function, giving the Cohen-Macaulay case.

Source reference. Original reference 0F91: sources/stacks-chow/pinned-original-source.tex, source lines 1433–1433. Navigation pointer for \ref{example-CM-irreducible} used inside the counted span (line 8311); the argument that consumes it is retained in the span.

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Source reference. Original lemma 0FIW: sources/stacks-chow/tags.txt, source lines 19406–19406. Navigation pointer for \ref{local-cohomology-lemma-catenary-S2-equidimensional} used inside the counted span (line 8296); the argument that consumes it is retained in the span.

Source reference. Original lemma 033N: sources/stacks-chow/tags.txt, source lines 4062–4062. Navigation pointer for \ref{properties-lemma-normal-locally-Noetherian} used inside the counted span (line 8307); the argument that consumes it is retained in the span.

Source reference. Original section 02QK: sources/stacks-chow/pinned-original-source.tex, source lines 1357–1357. Navigation pointer for \ref{section-setup} used inside the counted span (line 8280); the argument that consumes it is retained in the span.

Source reference. Original reference 02QL: sources/stacks-chow/pinned-original-source.tex, source lines 1372–1372. Navigation pointer for \ref{situation-setup} used inside the counted span (line 8278); the argument that consumes it is retained in the span.

Source reference. Original lemma 02IB: sources/stacks-chow/tags.txt, source lines 3294–3294. Navigation pointer for \ref{topology-lemma-dimension-function-unique} used inside the counted span (line 8313); the argument that consumes it is retained in the span.

Lemma 0.1. *Let (S, δ) be as in Situation 02QL. Let X be locally of finite type over S. Write $\delta = \delta_{X/S}$ as in Section 02QK. The following are equivalent*

1. *There exists a decomposition $X = \coprod_{n \in \mathbf{Z}} X_n$ into open and closed subschemes such that $\delta(\xi) = n$ whenever $\xi \in X_n$ is a generic point of an irreducible component of X_n.*
2. *For all $x \in X$ there exists an open neighbourhood $U \subset X$ of x and an integer n such that $\delta(\xi) = n$ whenever $\xi \in U$ is a generic point of an irreducible component of U.*
3. *For all $x \in X$ there exists an integer n_x such that $\delta(\xi) = n_x$ for any generic point ξ of an irreducible component of X containing x.*

The conditions are satisfied if X is either normal or Cohen-Macaulay[1]*.*

Proof. It is clear that (1) $\Rightarrow$ (2) $\Rightarrow$ (3). Conversely, if (3) holds, then we set $X_n = \{x \in X \mid n_x = n\}$ and we get a decomposition as in (1). Namely, X_n is open because given x the union of the irreducible components of X passing through x minus the union of the irreducible components of X not passing through x is an open neighbourhood of x. If X is normal, then X is a disjoint union of integral schemes (Properties, Lemma 033N) and hence the properties hold. If X is Cohen-Macaulay, then $\delta' : X \to \mathbf{Z}$, $x \mapsto -\dim(\mathcal{O}_{X,x})$ is a dimension function on X (see Example 0F91). Since $\delta - \delta'$ is locally constant (Topology, Lemma 02IB) and since $\delta'(\xi) = 0$ for every generic point ξ of X we see that (2) holds. □

[1] In fact, it suffices if X is (S_2). Compare with Local Cohomology, Lemma 0FIW.